H.T.No: | | | | | | | | | | |

USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY : NG

(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)

II B.Tech II Semester Regular Examinations, April-2025, Regulation: UR23

23HM401A: MANAGERIAL ECONOMICS & FINANCIAL ANALYSIS

(EEE/ ECE/ CSE/ IT/ AI&DS/ AI&ML)

Time: 3 hours **Max. Marks: 70**

PART-A Compulsory Questions (Each carry 2 Marks) **10 X 2 =20M**

I.	**a)**	Define Income elasticity of demand?		**(BL1)**
	b)	What is meant by Macroeconomics?		**(BL1)**
	c)	What are the variable factors used in production?		**(BL1)**
	d)	Define break even analysis		**(BL1)**
	e)	What is meant by business organization?		**(BL1)**
	f)	Define imperfect market?		**(BL1)**
	g)	How do you estimate the working capital		**(BL1)**
	h)	What meant by payback period?		**(BL1)**
	i)	What is meant by double entry book keeping?		**(BL1)**
	j)	What is meant by balance sheet?		**(BL1)**

PART-B (Answer either A or B from each Question) **5 X 10 =50M**

II.A.	**1)**	Illustrate the Law of Demand with assumptions and exceptions with the help of Individual demand schedule and demand curve.	**10M**	**(BL4)**
		OR		
II.B.	**1)**	How managerial economics is related to other subjects?	**10M**	**(BL1)**
III.A.	**1)**	Explain the law of returns to scale in long run production?	**10M**	**(BL1)**
		OR		
III.B.	**1)**	Explain the advantages of cost analysis?	**10M**	**(BL1)**
IV.A.	**1)**	Explain the features of monopolistic competition?	**5M**	**(BL1)**
	2)	Explain the different types of partners?	**5 M**	**(BL1)**
		OR		
IV.B.	**1)**	Explain the pricing strategies with examples?	**10M**	**(BL1)**
V.A.	**1)**	Explain traditional methods of capital budgeting?	**5M**	**(BL1)**

	2)	Explain the Modern methods of capital?	**5M**	**(BL1)**

OR

V.B.	**1)**	Explain the Sources of Capital budgeting?	**10M**	**(BL1)**
VI.A.	**1)**	Explain the importance of journal book in accounting?	**5M**	**(BL1)**
	2)	Explain the importance of double entry system?	**5M**	**(BL1)**

OR

VI .B.	**1)**	Explain the Activity ratio Ratio with formula?	**5M**	**(BL1)**
	2)	Explain the Capital Structure Ratio?	**5M**	**(BL1)**

H.T.No: | | | | | | | | | | |

USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY : NG
(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)
II B.Tech II Semester Regular Examinations, April-2025, Regulation: UR23
23ES402C: PROBABILITY & STATISTICS
(Common to CSE, IT)

Time: 3 hours **Max. Marks: 70**

PART-A Compulsory Questions (Each carry 2 Marks) **10 X 2 =20M**

I. a) What are the merits demerits of arithmetic mean. **(BL2)**
b) Define Skewness **(BL2)**
c) Write Normal equations for Parabola **(BL4)**
d) Define Regression **(BL4)**
e) State Baye's Theorem **(BL3)**
f) Define Normal distribution **(BL3)**
g) If mean $\bar{x} = 31.45,\ \mu = 32.3, \sigma = 1.6\ \&\ n = 40$ find z - test Statistic **(BL6)**
h) Define Point estimator **(BL6)**
i) Define Null Hypothesis **(BL4)**
j) Explain the term two Tailed test **(BL4)**

PART-B (Answer either A or B from each Question) **5 X 10 =50M**

II.A. 1) Calculate the median for the following frequency distribution: **5M (BL2)**

x	1	2	3	4	5	6	7	8	9
f	8	10	11	16	20	25	15	9	6

2) Calculate Quartile deviation (Q.D.) from mean for the following data: **5M (BL2)**

Marks	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 - 70
No. of students	6	5	8	15	7	6	3

OR

II.B. 1) Compute the geometric mean from the following data **5M (BL2)**

Marks	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
No. of students	10	5	8	7	20

2) An incomplete frequency distribution demand as table **5M (BL2)**

Variables	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80
Frequency	12	30	….	65	…..	25	18

Given that median is 46 and N=229.Determine the missing frequencies using the median formula.

III.A.1) By the method of least squares, find the straight line that best fit to the following data **5M (BL4)**

X	1	2	3	4	5
Y	2	5	4	7	10

2) The ranks of same 16 students in Mathematics and Physics are as follows. Two numbers within brackets denote the ranks of the students in Mathematics and Physics: (1,1) (2,10) (3,3) (4,4) (5,5) (6,7) (7,2) (8,6) (9,8) (10,11) (11,15) (12,9) (13,14) (14,12) (15,16) (16,13).Calculate the rank Correlation Coefficient for proficiencies of this group in Mathematics and Physics **5M (BL4)**

OR

III.B.1) Fit a curve of the form $y = ab^x$ **5M (BL4)**

x	0	1	2	3	4	5	6	7
y	10	21	35	59	92	200	400	610

2) Find the most likely Price in Mumbai corresponding to the Price of 70 at Kolkata from the following **5M (BL4)**

	Kolkata	Mumbai
Average Price	65	67
S. D	2.5	3.5

Correlation coefficient between Prices of commodities in the two Cities is 0.8

IV.A.1) For the continuous probability function $f(x) = kx^2e^{-x}, x \geq 0$. Find (i) k, (ii) Mean, (iii)Variance iv) SD **10M (BL3)**

OR

IV.B.1) An irregular six-faced die is thrown and the expectation that in 10 throws it will give five even numbers is twice the expectation that it will give four even numbers. How many items in 10,000 sets of 10 throws each, would you expect it to give no even number? **5M (BL3)**

2) In a distribution exactly normal, 10.03% of the items are under 25 kilogram weight and 89.97% of the items are under 70 kilogram weigh. What are the mean and standard deviations of the distribution? **5M (BL3)**

V.A. 1) Samples of size 2 are taken from the population 1, 2, 3, 4, 5, 6 **with replacement**. Find **10M (BL6)**
(i) Population mean
(ii)Population standard deviation
(iii) Mean of the sampling distribution of means (iv) Standard deviation of the sampling distribution of mean

OR

V.B. 1) A random sample of size 64 is taken from a normal population with mean μ=51.4 and σ=6.8. What is the probability that the mean of the sample will **5M (BL6)**

i) Exceed 52.9

Fall between 50.5 and 52.3

2) A normal population has a mean of 0.1 and SD of 2.1. Find the probability that mean of a sample of size 900 will be negative. **5M (BL6)**

VI.A.1) It is claimed that a random sample of 49 tyres has a mean life of 15200 km. This sample was drawn from a population whose mean is 15150 km and standard deviation of 1200 km. Test the significance at 0.05 level. **5M (BL4)**

2) Random samples of 400 men and 600 women were asked whether they would like to have a flyover near their residence. 200 men and 325 women were in favour of the proposal. Test the hypothesis that proportions of men and women in favour of the proposal are same against that they are not, at 5% level. **5M (BL4)**

OR

VI.B. 1) The heights of 10 males of a give locality are found to be 70,67,62,68,61,68,70,64,64,66 inches. Is it reasonable to believe that the average height is greater than 64 inches? **5M (BL4)**

2) A Sample analysis of examination results of 500 students was made. It was found that 220 students had failed, 170 had secured a third class, 90 were placed in second class and 20 got a first class. Do these figures commensurate with the general examination result which is in the ratio 4 : 3 : 2 : 1 for the various categories respectively. **5M (BL4)**

H.T.No: | | | | | | | | | |

USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY : NG

(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)

II B.Tech II Semester Regular Examinations, April-2025, Regulation: UR23

23PCCS403: DATA WAREHOUSING AND DATA MINING

(CSE/ AI&DS/ AI&ML)

Time: 3 hours **Max. Marks: 70**

PART-A Compulsory Questions (Each carry 2 Marks) **10 X 2 =20M**

I.	**a)**	What is a data ware house.	**(BL 1)**
	b)	Define virtual warehouse.	**(BL 1)**
	c)	List the types of data ?	**(BL 2)**
	d)	Define Data quality	**(BL 1)**
	e)	What is the purpose of the holdout method in classification?	**(BL 2)**
	f)	Define Cross Validation.	**(BL 2)**
	g)	What is the Apriori principle?	**(BL 2)**
	h)	What is the FP-Growth algorithm?	**(BL 1)**
	i)	What are the two main types of clustering methods?	**(BL 1)**
	j)	What is the objective of the K-means algorithm?	**(BL 2)**

PART-B (Answer either A or B from each Question) **5 X 10 =50M**

II.A.	**1)**	Compare Operational Database Systems and Data Warehouses.	**5M**	**(BL 2)**
	2)	Write short note on OLAP systems.	**5M**	**(BL2)**
		OR		
II.B.	**1)**	Illustrate bitmap indexing OLAP data in Data Warehouse implementation.	**5M**	**(BL 2)**
	2)	Demonstrate a fact constellation schema in Data Warehouse.	**5M**	**(BL 2)**
III.A.	**1)**	Illustrate data mining tasks with examples.	**10M**	**(BL 1)**
		OR		
III.B.	**1)**	Illustrate the terms types of data, attribute and measurement.	**5M**	**(BL 1)**
	2)	Elaborate different types of attributes with example and operations.	**5M**	**(BL 5)**
IV.A.	**1)**	Explain Naïve Bayes Classifier? How to use Bayes theorem for classification.	**10M**	**(BL 3)**
		OR		
IV.B.	**1)**	How to build a decision tree? Explain briefly about Hunts algorithm.	**5M**	**(BL 3)**
	2)	Illustrate Bayesian belief network classification with an example.	**5M**	**(BL 3)**
V.A.	**1)**	Write about problem definition in association analysis.	**10M**	**(BL 2)**
		OR		
V.B.	**1)**	Classify the compact representation of frequent items.	**10M**	**(BL 2)**
VI.A.	**1)**	Classify Classical Partitioning Methods	**10M**	**(BL 4)**
		OR		
VI .B.	**1)**	Justify Whether the purpose of the clustering is utility.	**5M**	**(BL 5)**
	2)	What is a Cluster analysis and explain different ways of Clustering to same set of Points.	**5M**	**(BL 2)**

H.T.No: | | | | | | | | | | |

USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY : NG

(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)

II B.Tech II Semester Regular Examinations, April-2025, Regulation: UR23

23PCCS404 : OPERATING SYSTEM

(CSE/ IT)

Time: 3 hours **Max. Marks: 70**

PART-A Compulsory Questions (Each carry 2 Marks) **10 X 2 =20M**

I.	**a)**	Identify the difference between mainframe and desktop operating system.	**(BL2)**
	b)	What is real time system?	**(BL1)**
	c)	How will you calculate turn-around time?	**(BL2)**
	d)	Difference between Process and Program	**(BL2)**
	e)	What is a semaphore?	**(BL1)**
	f)	What are the conditions under which a deadlock situation may arise?	**(BL1)**
	g)	Define swapping.	**(BL1)**
	h)	What is Belady's anomaly?	**(BL1)**
	i)	Define seek time and latency time.	**(BL1)**
	j)	Define Equal Allocation.	**(BL1)**

PART-B (Answer either A or B from each Question) **5 X 10 =50M**

II.A.	**1)**	Discuss in detail the Operating System Functions.	**10M**	**(BL2)**
		OR		
II.B.	**1)**	Discuss with help of a neat diagram the structure of Operating System.	**10M**	**(BL2)**
III.A.	**1)**	Discuss about process scheduling based on queues with diagram	**10M**	**(BL2)**
		OR		
III.B.	**1)**	Explain and differentiate between user level and kernel level thread.	**10M**	**(BL2)**
IV.A.	**1)**	Describe about semaphores and its roles synchronization.	**10M**	**(BL2)**
		OR		
IV.B.	**1)**	Describe the Readers and writers' problem. How to handle it?	**10M**	**(BL2)**
V.A.	**1)**	Given page reference string: 1,2,3,2,1,5,2,1,6,2,5,6,3,1,3,6,1,2,4,3. Assuming there are 3 frames Compare the number of page faults for FIFO and MRU page replacement algorithm.	**10M**	**(BL4)**
		OR		
V.B.	**1)**	Describe with examples the page replacement algorithms.	**10M**	**(BL2)**
VI.A.	**1)**	Describe the C-SCAN,LOOK and C-LOOK disk scheduling algorithm using the following data. The dist head is initially at position-cylinder 53.the cylinder sequence of requests is 98, 183, 37, 122, 14, 124, 65, 67. Find the total head movement.	**10M**	**(BL6)**
		OR		
VI .B.	**1)**	Explain file allocation methods in detail.	**10M**	**(BL1)**

H.T.No: | | | | | | | | | | |

USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY : NG

(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)

II B.Tech II Semester Regular Examinations, April-2025, Regulation: UR23

23PCCS405: SOFTWARE ENGINEERING

(CSE/ IT)

Time: 3 hours **Max. Marks: 70**

PART-A Compulsory Questions (Each carry 2 Marks) **10 X 2 =20M**

I.	**a)**	List different software life cycle models?	**(BL2)**
	b)	Write the difference between system software and application software?	**(BL1)**
	c)	What are the Metrics for project size estimation?	**(BL2)**
	d)	List the users of SRS?	**(BL1)**
	e)	Define DFD?	**(BL2)**
	f)	What is agility in software development?	**(BL1)**
	g)	What is white-box testing?	**(BL2)**
	h)	What is debugging?	**(BL2)**
	i)	List the different types of CASE tools?.	**(BL2)**
	j)	List the software maintenance process models?	**(BL2)**

PART-B (Answer either A or B from each Question) **5 X 10 =50M**

II.A.	**1)**	Summarize the evolution of Software Engineering, highlighting major milestones and advancements over time.	**10M**	**(BL2)**
		OR		
II.B.	**1)**	Compare and contrast the Waterfall, Spiral, and Agile models of software development. Provide examples of situations where each model is suitable.	**10M**	**(BL4)**
III.A.	**1)**	Analyze the various complexities involved in Software Project Management.	**10M**	**(BL4)**
		OR		
III.B.	**1)**	Differentiate between Single Variable and Multivariable Heuristic Estimation Techniques and analyze their applicability using examples.	**10M**	**(BL4)**
IV.A.	**1)**	Define cohesion in software design and illustrate various types of cohesion using suitable examples.	**10M**	**(BL3)**
		OR		
IV.B.	**1)**	Compare and contrast function-oriented and object-oriented software design approaches using appropriate examples.	**10M**	**(BL4)**
V.A.	**1)**	Explain the coding phase in software engineering. What are the objectives, standards, and guidelines involved?	**10M**	**(BL3)**
		OR		
V.B.	**1)**	Discuss about Software Quality management system.	**10M**	**(BL3)**
VI.A.	**1)**	Explain the architecture of a CASE environment and describe its major components and their roles.	**10M**	**(BL2)**
		OR		
VI .B.	**1)**	Describe the four common software maintenance models. Highlight how they differ in approach and application.	**10M**	**(BL3)**